

EDUCATION

- **McGill University** Montreal, Canada
Ph.D. in Neuroscience, Supervisor: Prof. Gustavo Turecki Jan 2027 – Expected 2031
- **Wuhan University** Wuhan, China
M.Med. in Psychiatry and Mental Health, Supervisor: Prof. Zhongchun Liu Sep 2023 – Jun 2026
- **Central South University** Changsha, China
B.M., Clinical Medicine Sep 2018 – Jun 2023

SELECTED PUBLICATIONS

- Zhang L**, Duan H, Lyu H, Ma S, Xu Y, Xie X, Gong Q, Kang L, Chen S, Liu Z. Multi-omics analysis reveals CXCL14+ inhibitory neuron dysfunction in major depressive disorder. *J Affect Disord.* 2026;405:121586. [\[Paper\]](#)
- Zhang L**, Chen X, Xu Y, Liu J, Liu Z. Exploring glucagon-like peptide-1 receptor agonists as potential disease-modifying agent in psychiatric and neurodevelopmental conditions: Evidence from a drug target Mendelian randomization. *BMC Psychiatry.* 2025;25(1):484. [\[Paper\]](#)
- Feng Y, Li N, **Zhang L**, Wang W, Duan H, Sun S, Li R, Zhang Y, Song X, Zhang Y, Zhang H, Nie Z, Yan H, Wang C, Liu Z. X-linked USP11 drives depression-like behaviors by stabilizing CK2 α and disrupting mitochondrial function. *CNS Neurosci Ther.* 2026;32(6):e70934. [\[Paper\]](#)
- Gong Q, Lyu H, Kang L, Ma S, Zhang N, Xie XH, Zhou E, Deng Z, **Zhang L**, Liu J, Liu Z. The genetic relationships between post-traumatic stress disorder and its corresponding neural circuit structures. *J Affect Disord.* 2026;393:120323. [\[Paper\]](#)
- Lyu H, Kang L, Gong Q, Xie XH, Ma S, Yao L, Chen MM, **Zhang L**, Yu H, Wang X, Wang C, Liu Z. The relationship between vitamin D levels and depression: A genetically informed study. *Nutr J.* 2025;24(1):158. [\[Paper\]](#)
- Yuan R, Yang Y, Wang Z, Song Z, **Zhang L**, Guo Z, Yu H, Ju Y, Feng Z. Adolescent psychological resilience and non-suicidal self-injury: A moderated mediation model. *Chin J Clin Psychol.* 2024;32(4):843–849. [\[Paper\]](#)
- Zhang L**, Xu Y, Ma Y, Xie T, Liu C, Liu Q. Research trends in transient receptor potential vanilloid in cardiovascular disease: Bibliometric analysis and visualization. *Front Cardiovasc Med.* 2023;10:1071198. [\[Paper\]](#)

ACADEMIC EXPERIENCES

- **A Multidimensional Plasma Proteomic Atlas of Depression** Fudan University
Supervisor: Prof. Wei Cheng, Institute of Science and Technology for Brain-Inspired Intelligence Sep 2025 – Jan 2026
 - Analyzed large-scale plasma proteomics data from the GNPC discovery cohort and the UK Biobank validation cohort to characterize depression-associated molecular signatures.
 - Identified and externally validated robust protein markers associated with depression status, symptom severity, and incident depression.
 - Developed a proteomic risk score for prediction and identified three biologically distinct depression subtypes through unsupervised clustering.
- **Investigating Neuronal Dysfunction in Major Depressive Disorder Through Multi-Omics Integration** Wuhan University
Supervisor: Prof. Zhongchun Liu, Department of Psychiatry, Renmin Hospital of Wuhan University Sep 2023 – June 2026
 - Integrated multi-omics datasets including scRNA-seq, GWAS summary statistics, and UK Biobank plasma proteomics to identify 273 MDD-associated genes and prioritize MDD-relevant subtypes.
 - Defined meta-programs in excitatory and inhibitory neurons by non-negative matrix factorization, uncovering transcriptional modules linked to synaptic plasticity, neuronal connectivity, and epigenetic regulation; conducted trajectory analysis to reveal pathological states in neuronal subtypes.
 - Characterized a *CXCL14*⁺ inhibitory neuron subtype involved in stress response and cell-cell communication; validated 26 risk genes through proteomics, with 5 protein-coding genes predictive of patient prognosis using CoxBoost regression.
- **Exploring glucagon-like peptide-1 receptor agonists as potential disease-modifying agents in psychiatric and neurodevelopmental conditions** Wuhan University
Supervisor: Prof. Zhongchun Liu, Department of Psychiatry, Renmin Hospital of Wuhan University Sep 2023 – Sep 2024

- Conducted Mendelian randomization analysis to evaluate the potential effects of GLP-1 receptor agonists on 12 psychiatric and neurodevelopmental disorders.
 - Integrated cis-eQTL and large-scale GWAS data from FinnGen, PGC, and UK Biobank; performed meta-analysis and sensitivity testing.
 - Identified risk-lowering associations with schizophrenia, PTSD, autism, and others, with an opposing association for OCD.
- Excellent Doctor Training Program** Central South University
Supervisor: Prof. Qiming Liu, Department of Cardiology, the Second Xiangya Hospital Mar 2021 – Jun 2023
 - Participated in outpatient consultations and inpatient clinical practice under Prof. Qiming Liu's supervision in the Cardiology II Department; observed over ten catheter ablation procedures.
 - Developed proficiency in diagnosing and managing common cardiovascular diseases, as well as foundational skills in both wet-lab and bioinformatics methods.
 - Studied key interdisciplinary topics including advanced manufacturing technologies, biomaterials and biosensors, computational modeling, machine learning, and medical image segmentation.
- Exploring the Relationship Between Adolescent Psychological Resilience and Non-Suicidal Self-Injury (NSSI): The Mediating Effect of Loneliness.** Central South University
Supervisor: Prof. Yan Zhang, National Center for Mental Disorders, the Second Xiangya Hospital Mar 2021 – Mar 2022
 - Led a school-based study in Changsha examining whether psychological resilience protects against adolescent NSSI; designed protocols, localized validated scales, and coordinated data collection with local stakeholders.
 - Recruited 1,100 participants and ensured data quality for 1,011 valid responses; conducted statistical analyses to examine mediation and moderation effects in adolescent NSSI. Using SPSS 26.0 to conduct statistical analyses including common method bias testing, correlation analysis, and binary logistic regression.
 - Found a 22.35% NSSI prevalence; identified depression, anxiety, residence, loneliness, sleep problems, and low resilience as significant risk factors. Established a moderated mediation model showing sleep issues amplify loneliness's impact on NSSI and informed prevention strategies.
- Exploring the Role and Mechanisms of Wnt1 in Postmenopausal Osteoporosis** Central South University
Supervisor: Prof. Tianlong Huang, the Second Xiangya Hospital Mar 2020 – Mar 2021
 - Developed an in vitro model of postmenopausal osteoporosis using MC3T3 cells and investigated the expression of the non-canonical Wnt1 signaling pathway.
 - Validated Wnt1 overexpression and its involvement in the mTORC1 pathway using molecular assays.

SCHOLARLY CONTRIBUTION

Contributed to the development of *the 2025 Chinese Guidelines for the Diagnosis and Treatment of Bipolar Disorder* under the supervision of Prof. Zhongchun Liu. Responsible for drafting the background section, conducting literature reviews, integrating clinical evidence, and formulating treatment recommendations. Acknowledged for contributions in the final published guideline.

LAB EXPERIENCE

- Institute of Science and Technology for Brain Inspired Intelligence** Fudan University
Visiting Student Sep 2025 – Jan 2026
 - Conducted research in computational psychiatry under the supervision of Prof. Wei Cheng, with a focus on depression related plasma proteomics and multi-omics integration.
 - Analyzed GNPC and UK Biobank cohorts to identify molecular signatures of depression, construct predictive models, and define biologically meaningful subtypes.
- National Center for Mental Health, the Second Xiangya Hospital** Central South University
Research Assistant Mar 2022 – Jun 2023
 - Supported clinical research by contributing to study design and statistical analysis.
 - Performed data management and outcome analysis using R and SPSS.
- Hunan Institute of Cardiovascular Disease, the Second Xiangya Hospital** Central South University
Research Assistant Mar 2021 – Mar 2022
 - Trained in the use of the Langendorff system for establishing an ex vivo cardiac perfusion model in mice.
 - Performed protein analysis of cardiomyocytes using techniques such as Western blotting and qPCR.

CONFERENCES & PRESENTATIONS

- **The 22nd Annual Conference of the Chinese Psychiatrist Association** Tianjin, China
Poster Presentation July 2026
 - **Title:** *Mapping Transdiagnostic Symptom Dimensions to Brain Cell-State Programs in Major Psychiatric Disorders*

HONORS & AWARDS

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| McGill-CSC Joint Scholarship | <i>China Scholarship Council, 2026</i> |
| Outstanding Graduate Student Award | <i>Wuhan University, 2025</i> |
| First Prize Scholarship | <i>Wuhan University, 2025</i> |
| Outstanding Social Practice Team Award | <i>Central South University, 2021</i> |
| Outstanding Student Award | <i>Central South University, 2019</i> |
| First Prize Scholarship | <i>Central South University, 2019</i> |
| Zhangyuan Tungsten Industry Scholarship | <i>Hongyi Zhangyuan Tungsten Co., Ltd & Central South University, 2019</i> |

SPECIALIZED SKILLS & INTERESTS

- **Programming & Data Analysis:** R, Python, bash, LaTeX; PyTorch, TensorFlow, scikit-learn, pandas, NumPy.
- **Bioinformatics Tools & Pipelines:** Seurat, Scanpy, ArchR, cNMF, Cell Ranger, Signac, PLINK, Snakemake.
- **Experimental Techniques:** Western blotting, qPCR, immunofluorescence, cell culture, animal handling.
- **Data Visualization:** ggplot2, Matplotlib, Seaborn.
- **Languages:** Mandarin (native); English (IELTS Academic: Overall Band 6.5).
- **Research Interests:** Computational psychiatry, psychiatric genomics, single-cell and spatial omics, and multi-omics integration.